

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Water treatment, conservation & new sources - Set 1

Mark scheme

AQA spec 3.2.2.4-3.2.2.5; The physical environment; textbook pp. 122-127

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Topography: a narrow exit from a large deep basin so a small dam holds a large volume (1) • Geology: impermeable rock beneath (no seepage), strong enough to support the dam, no faults / seismic activity (1) • Catchment area: large enough with regular, reliable rainfall / river inflow; climate not too hot (evaporation losses) (1) • Catchment land use: low pollution risk (industry, pesticides); low soil erosion (sedimentation reduces capacity) (1) • Infrastructure / proximity to demand: access for construction, cost of transporting water (1) • Existing land use: value of what is lost (urban areas, conservation sites, farmland) - loss:benefit analysis (1)	4
01.2	<i>Up to 3 marks in total; award marks in proportion to the completeness of each point.</i> • Advantage: fewer land-use conflicts as no land is flooded; large volume near coastal cities (up to 2) • Disadvantages: destroys / changes important intertidal habitats (mudflats for wading birds); obstacle to shipping; collects pollution from the whole river catchment; sedimentation (up to 2)	3
01.3	<i>1 mark for each point.</i> • Very energy-intensive and expensive (reverse osmosis / distillation) (1) • Only worthwhile where seawater is available and freshwater supplies are inadequate (eg Malta, Middle East) (1)	2

Question	Marking guidance	Marks
01.4	<i>1 mark.</i> Surplus water in wet periods (when infiltration capacity is exceeded and runoff would go to the sea) is pumped underground or diverted into lagoons to infiltrate, so abstraction in dry periods is balanced by recharge (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-124; AQA spec 3.2.2.5.	
02.1	<i>1 mark for the correct answer.</i> $(150 - 125) / 150 \times 100 = 16.7\%$ (accept 17%) (1)	1
02.2	<i>1 mark for each point.</i> - A fixed fee gives no incentive to save water because the bill is the same however much is used (1) - Metering matches the charge to the volume used so households save money by using less (fewer hosepipes, low-flow appliances) (1)	2
02.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Low-water appliances: dual-flush toilets, low-flow shower heads, efficient washing machines / dishwashers, tap timers (1) - Greywater re-use: bath / shower water used to flush toilets or water gardens (1) - Rainwater collection / water butts for garden use (1) - Xeriscaping: planting drought-adapted species that need no watering (1) - Repairing leaks / dripping taps (1)	3
02.4	<i>1 mark for each point.</i> - Old infrastructure / pipes in poor condition leak (1) - Repair and good maintenance / pipe replacement / leak detection reduce wastage (1)	2
02.5	<i>1 mark for each point.</i> - Water is delivered directly to the plant roots / soil around each plant (1) - Less evaporation loss from spray droplets and from wetting the whole soil surface; less runoff (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 124-126; AQA spec 3.2.2.5.	
03.1	<i>1 mark for each point.</i> - Clay particles are very small and carry electrostatic charges on their surfaces so they repel each other and stay suspended (1) - Flocculants (aluminium sulfate / alum or polyelectrolytes) are added and mixed to neutralise the charges (1) - The particles clump together (flocs) and settle in the clarifier tank (1)	3
03.2	<i>1 mark for each point.</i> - Increases dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from anaerobic sources (1) - Oxidises some dissolved metals (eg iron) so they become insoluble and can be removed; improves taste (1)	2
03.3	<i>1 mark for each point.</i> - Peaty water contains organic matter which reacts with chlorine to produce toxic substances (eg trihalomethanes) (1) - Ozone / UV kill pathogens without forming these by-products (though chlorine / chloramine keeps water sterile during distribution) (1)	2

Question	Marking guidance	Marks
03.4	<i>1 mark for each point.</i> • Ion exchange: toxic ions adsorb onto polymer resin beads or zeolite (also used to soften water by removing calcium / magnesium) (1) • Reverse osmosis / distillation (desalination) - expensive, energy-intensive (1) • Activated carbon: organic molecules causing odour / taste and pesticides adsorb onto the high-surface-area carbon particles (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-128; AQA spec 3.2.2.4.	
04.1	<i>Correct answer only.</i> • B - rivers are accessible over a large area with few natural contaminants (1)	1
04.2	<i>1 mark for each point.</i> • Potable / public supply: no pathogens, low toxins, good appearance / taste / smell (public health) (1) • Industrial boiler / cooling water: soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange; spray irrigation: low turbidity to avoid blocked pipes and low heavy metals (1)	2
04.3	<i>1 mark for each point.</i> • Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) (1) • Disadvantage: high cost / energy for pumping; ecological impacts on the donor river; transfer of pollutants or invasive species; political conflict (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
05.1	<i>Correct answer only.</i> • Its total annual discharge (1)	1
05.2	<i>1 mark for each point.</i> • Public supplies cannot match consumer demand / the water is often cleaner than rivers (1) • It also reduces urban flooding after heavy rain; surplus can be used for aquifer recharge (1)	2
05.3	<i>1 mark for each point.</i> • The reservoir is created in the estuary so no farmland / settlements need to be flooded (1) • But it changes important inter-tidal habitats, obstructs shipping and can be polluted by activities anywhere in the river's catchment (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-123; AQA spec 3.2.2.5.	
06.1	<i>1 mark for each point.</i> • Potable: no pathogens, toxins at acceptable levels, looks / smells / tastes good - public health, consumer complaints (1) • Spray irrigation: low turbidity, low toxins such as heavy metals - blockage of pipes, contamination of food (1) • Condenser cooling: no gross solids - pipe blockage (1) • Boiler water: no dissolved minerals / soft water - mineral deposits reduce heat exchange and block pipes (1)	4

Question	Marking guidance	Marks
06.2	<i>1 mark for each point.</i> • Raises dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from decay of organic matter (1) • Some dissolved toxic / bad-tasting metals become insoluble when oxidised and can be removed (1)	2
06.3	<i>1 mark for each point.</i> • Clay particles carry electrostatic charges that make them repel each other so they stay suspended (1) • Flocculants such as aluminium sulfate (alum) or polyelectrolytes neutralise the charges so the particles clump and settle in the clarifier (1)	2
06.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Chlorine (1) • Ozone (1) • Ultraviolet light (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-127; AQA spec 3.2.2.4.	
07.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • No reservoir range = $3.9 - 0.7 = 3.2$ m; with reservoir = $2.7 - 1.7 = 1.0$ m (1) • Reduction = 2.2 m (1)	2
07.2	<i>1 mark for each point.</i> • After heavy rainfall surplus water is stored, reducing peak flows / flooding downstream (1) • During dry periods the dam is opened to release water and maintain river levels (1)	2
07.3	<i>1 mark for each point.</i> • Deforestation / urbanisation reduce infiltration and interception (1) • So runoff reaches the river faster after rain and less water is stored to maintain flow in dry periods (1)	2

Question	Marking guidance	Marks
07.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Site factors: topography - narrow exit from a large deep basin so a small dam holds a large volume; impermeable, strong bedrock without faults or seismic activity; large catchment with regular high rainfall and low evaporation; catchment land use with low pollution risk (pesticides, industry) and low soil erosion to avoid sedimentation; forest / vegetation in the flooded area decaying anaerobically releasing methane; access for workers and materials and distance to demand; value of the land lost (urban, conservation, agricultural) - loss-benefit analysis differs between countries • Alternatives for supply: rainwater collection, rivers, estuary barrages, desalination (energy-intensive so only where freshwater is inadequate), unexploited aquifers (N Africa, Kenya), inter-basin transfers (China, Australia, Wales-England), artificial aquifer recharge using surplus wet-season water, river-regulation reservoirs, afforestation to even out flow • Conservation strategies: low-water appliances, dual-flush toilets, xeriscaping, drip irrigation, grey-water re-use, leak repair (about 20% lost in the UK), metering • Treatment / pollution control allowing re-use of contaminated sources: sewage treatment, buffer strips, mine drainage control • Evaluation: reservoirs give large reliable supplies and flood control but flood land, cause habitat loss and methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination has high energy / carbon cost; combination needed as demand grows • Justified conclusion	9
	Model answer - Level 3 (8 marks)	
	A reservoir site is judged first on topography: a narrow exit from a large, deep basin lets a small, cheap dam hold a large volume. The bedrock must be impermeable so water is not lost and strong enough, without faults or seismic activity, to carry the dam. The catchment should be large with regular, high rainfall and low evaporation, and its land uses should not pose a pollution risk from industry or pesticides or cause the soil erosion that would silt the reservoir. Flooding forest or vegetation causes anaerobic decay and methane. Access for workers and materials, distance to the area of demand, and the value of the land lost - urban areas and nature reserves protected, farmland less so - complete a loss-benefit analysis that different countries weigh differently.	

Question	Marking guidance	Marks
	Supply can also be increased without new reservoirs: rainwater collection in cities, estuary barrages, desalination where freshwater is genuinely inadequate, newly found aquifers in North Africa and Kenya, inter-basin transfers as in China and from Wales to England, artificial recharge of aquifers with wet-season surplus, river-regulation reservoirs and afforestation to even out flow. Demand can be cut by low-water appliances, xeriscaping, drip irrigation, grey-water re-use, metering and repairing the leaks that lose about a fifth of UK supply, and treatment lets contaminated sources be re-used.	
	Reservoirs give large, reliable supplies and flood control but flood land and habitats and release methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination carries a high energy and carbon cost. As demand grows, the sustainable answer is conservation first, then a combination of sources matched to each region.	
	Weaker answer - Level 2 (4 marks)	
	A reservoir needs a large catchment area with lots of rain, impermeable rock so the water does not drain away, and a narrow valley so the dam is small. It should be near the people who need the water and not flood important land. Other ways to get water include desalination, which is expensive, and collecting rainwater. Water can be conserved by fixing leaks, using water meters and having low-flush toilets. All of these help make water supplies more sustainable.	
	What makes the difference	
	• Level 2 lists correct site factors but without the reasons (why impermeable, why the valley shape matters, why land use in the catchment matters, methane from flooded vegetation).	
	• Level 3 covers far more alternatives (aquifer recharge, river regulation, transfers, afforestation) and evaluates each; Level 2 evaluates only desalination.	
	• Level 3 ends with a prioritised judgement; Level 2 ends with 'all of these help'.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-127; AQA spec 3.2.2.4-3.2.2.5.	